JACQUELINE SIFFT

Bachelor of Science, Mathematics | MEi:CogSci Student

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RESEARCH INTERESTS

I'm motivated by how neuromodulation could both treat psychiatric disorders and teach us about the neural dynamics underlying them. I'm currently training toward that at the intersection of fMRI, computational modeling, and neuroscience, coming from a background in mathematics.

EXPERIENCE

Design and Implementation of Task-Based fMRI Paradigms for Studying TMS Treatment Effects in Depression (Research Project)

High Field MR Center, Center for Medical Physics and Biomedical Engineering, Medical University of Vienna, Austria

March 2026 – Ongoing — Vienna, Austria

- Literature review on neural networks/circuits, treatment types for major depressive disorder (MDD), resting-state and task fMRI and potential regions of interest (ROIs) for treatment response prediction; emphasis on fMRI-based MDD biomarkers, especially for transcranial magnetic stimulation (TMS).
- Practical introduction to TMS: Causing a motor-response by stimulating the left motor cortex.
- Designed and implemented a Go/No-go task paradigm and a novel fusion of implicit emotional differentiation and self-referential processing task with psychopy for use in a 3T MRI environment.

Spring School in Cognitive Modelling

University of Groningen

April 2026 — Groningen, Netherlands

- Lectures on Nengo-based neural modeling and neuromorphic intelligence
- Project: implemented distributed frequency encoding on a physical neuromorphic chip — exploring how populations of neurons can represent information beyond individual firing-rate limits.

Student Assistant for Studies

Center for Neuropediatrics, University Hospital, Rheinische Friedrich-Wilhelms-Universität Bonn.

2024 – 2025 — Bonn, Germany

- Administrative support for spinal muscular atrophy patients

Tutor for Algorithmic Mathematics

University of Bonn

2022 – 2023 — Bonn, Germany

- Led weekly exercise sessions in *algorithmic mathematics* (a combination of discrete mathematics and numerical methods), reinforcing concepts such as algorithm design, combinatorics, and computational techniques
- Presented and discussed solutions to assignments; addressed student questions
- Designed and facilitated in-class problem-solving exercises
- Graded weekly assignments and provided constructive feedback

EDUCATION

Master of Science, Cognitive Science

University of Vienna, Medical University of Vienna

October 2025 – Now — Vienna, Austria

fMRI, applications of fMRI, neuroscience, neuroinformatics, applied machine learning

Bachelor of Science, Mathematics

University of Bonn

2021 – 2025 — Bonn, Germany

Thesis: Whitehead Products (Algebraic Topology).

Exchange Semester, Mathematics

York University

2024 — Toronto, Canada

GoGlobal Scholarship. Data Science and Psychology, Linear Optimization.

TECHNICAL SKILLS

Programming, data & tools:
Python | SQL | Git | LaTeX | Linux

Neuroimaging & simulation:
Nilearn | Psychopy | FSL-MRS

LANGUAGES

German, native | English, C1

VOLUNTEERING

Equality Committee, Student Council Mathematics

University of Bonn

2021 – 2025 — Bonn, Germany

- Organized educational talks on mental health in mathematics and academia
- Organized community events for queer mathematicians, computer scientists and physicists

Global Ambassador

University of Bonn

2024 – 2025 — Canada and Germany

- Received training in intercultural communication
- Supported outreach events for the GoGlobal Program

PERSONAL INTERESTS

Rock Climbing | Fantasy Literature | Crocheting & Upcycling